

# Your Transformer is Secretly an EOT Solver

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ine months ago, or was it ten? I've learned not to trust my sense of time when deep in a problem. Anyone faced with a major deadline or enveloped in the mire of one particular question for an extended period intuitively understands that Time becomes something else then. Not the measured progression of hours, but rather a kind of thickness, a medium through which one moves with increasing difficulty, like a swimmer who has ventured too far from shore and finds the water has become viscous, resistant.

I was doing what I had seen others do, what perhaps I had been doing all along without quite admitting it to myself: trying to invent yet another subquadratic attention mechanism. Yes, another one. As if the world needed more of these things <sup>1</sup>. There are already so many. I knew this. You know this. Anyone who works in this field knows this. And yet we persist, as men have always persisted in rebuilding Babel, each generation convinced that this time, with these new tricks, with some *recherché* materials, with this understanding our predecessors lacked, the tower will stand.

But mine, of course, was going to be different. Mine was going to be new and quirky. Mine was going to reflect practical considerations; real constants and implementation details rather than mere Big-O algorithmic obscurantism.

The idea involved an approximation. This was to be an approximation that became asymptotically accurate as  $n$  (sequence length) grew. This way, it's only wrong in all the ways that don't matter and right in the one way that does. There was, naturally, some unanticipated problem. There is always some unanticipated problem. It's practically a theorem in this field.

## *Why Asymptotically Accurate Subquadratic Attention is Impossible*

Let me explain the nature of that unanticipated problem I mentioned. It is mathematically impossible for any subquadratic attention mechanism to be asymptotically accurate on all inputs.

The intuition is a *needle in a haystack* construction. The softmax function requires knowing all query-key similarities in each row to compute the normalization constant correctly. If an algorithm inspects only  $o(n^2)$  query-key pairs, then by the pigeonhole principle, there must exist at least one query row  $i^*$  for which it examines only  $o(n)$  of the  $n$  possible keys. Place an arbitrarily large similarity score in one of the uninspected positions, the "needle," and the algorithm will fail catastrophically because it lacks information essential to computing the attention weights.

More formally, define a subquadratic algorithm as one computing  $o(n^2)$  query-key dot products, and asymptotically accurate as producing output that converges to the exact attention matrix in the limit as  $n \rightarrow \infty$ . Assume, for contradiction, that such an algorithm  $\mathcal{A}$  exists.

By subquadraticity, the average number of keys inspected per query is  $o(n)$ . Therefore, there exists a query index  $i^*$  and a corresponding uninspected key index  $j^*$ . Now construct an adversarial input: let  $q_{i^*} = e_1$ , let  $k_{j^*} = M_n e_1$  where  $M_n = \ln(n^2)$ , and set all other queries, keys, and values to zero vectors. The exact attention output for row  $i^*$  concentrates almost entirely on position  $j^*$ , converging to  $e_1$  for  $n \rightarrow \infty$ . But since  $\mathcal{A}$  never inspects  $(i^*, j^*)$ , it has no information about this dominant contribution and must output approximately zero, yielding a non-vanishing error lower bound of  $\|e_1\|_2 = 1$  in the limit.

The contradiction arises from the global normalization inherent in softmax: computing accurate attention weights requires complete information about all query-key similarities in each row. A subquadratic algorithm cannot provide this complete information for all rows simultaneously, making asymptotic accuracy impossible.

So I walked away. I came back. I walked away again. And then, I had a thought. The thought was this: perhaps there is no way forward through brute force. Perhaps the only way through is to find some hidden mathematical structure, some symmetry or equivalence that's been sitting there all along, waiting for someone to notice it.

That thought, as it happens, was correct. Which is how I ended up writing this paper<sup>2</sup>. The paper, to be clear, has nothing to do with subquadratic attention. So I'm telling the story here: how trying to build something uninteresting led, through impossibility, to discovering something genuinely interesting: an exact equivalence between scaled dot-product attention and entropic optimal transport.

## *Attention as Entropic Optimal Transport*

We will demonstrate the capstone result of the paper: that the scaled dot-product attention weights for a collection of queries are equivalent to the unique solution of a particular one-sided entropic optimal transport problem. Specifically, one where the costs are given by the negative key-query dot products, the source space is equipped with the counting measure, and the target space is left unconstrained.

We begin by constructing this optimal transport problem. Consider the standard setup for computing attention weights for a set of  $n$  query vectors, collected rowwise into a real matrix of size  $n \times d_k$ .

$$Q = \begin{bmatrix} q_1^\top \\ \vdots \\ q_n^\top \end{bmatrix} \in \mathbb{R}^{n \times d_k}.$$

These attend over a collection of  $m$  key vectors collected rowwise into a real matrix of size  $m \times d_k$ .

$$K = \begin{bmatrix} k_1^\top \\ \vdots \\ k_m^\top \end{bmatrix} \in \mathbb{R}^{m \times d_k}.$$

Our objective is to formulate an optimal transport problem whose unique solution is exactly the attention weight matrix  $A \in \mathbb{R}_+^{n \times m}$ , where each component  $A_{ij}$  is defined by scaled dot-product attention with temperature  $\tau = \sqrt{d_k}$ :

$$A_{ij} \equiv \frac{\exp(q_i^\top k_j / \tau)}{\sum_{l=1}^m \exp(q_i^\top k_l / \tau)}.$$

Each row of  $A$  sums to one, so  $A$  is row-stochastic.

The formulation proceeds by specifying the source and target index sets, the measures on them, the feasible couplings, the cost, and the entropic regularization<sup>[3](#)</sup>.

### *Query source space and measure*

The source index set is  $1, 2, \dots, n$ , one index per query vector. The source measure  $a$  assigns unit mass to every source index. In other words,  $a_i = 1$  for all  $i$ , so the total source mass is  $n$ . You can think of each query  $q_i$  as emitting one unit of attention mass to be distributed across the target indices.

### *Key target space*

The target index set is  $1, 2, \dots, m$ , one index per key vector. Each  $j$  is a potential destination for mass originating at any source  $i$ .

A critical choice is how to treat the target marginal  $b$  on the target set. Unlike classical optimal transport that fixes both the source and target marginals, we leave the target marginal free. The distribution of mass arriving at the targets is not constrained ahead of time and will be determined by the optimization. This is a one-sided optimal transport problem.

## Transport plans and feasibility

A transport plan is a nonnegative matrix  $P \in \mathbb{R}_+^{n \times m}$ , where  $P_{ij}$  is the amount of mass sent from source  $i$  to target  $j$ . Feasibility is enforced only through the source constraints

$$\sum_{j=1}^m P_{ij} = a_i = 1$$

for all  $i \in \{1, \dots, n\}$ . There are no column constraints because the target marginal is free. The feasible set is therefore

$$U(a, \cdot) = \left\{ P \in \mathbb{R}_+^{n \times m} \mid \sum_{j=1}^m P_{ij} = 1 \right\}.$$

This row-sum constraint applies for all  $i$ . Each row of any feasible  $P$  lies in the standard probability simplex of dimension  $m - 1$ .

## Cost and entropy

We need a transport cost  $C_{ij}$  for moving mass from  $i$  to  $j$ . Attention prefers high similarity  $q_i^\top k_j$ , while optimal transport minimizes cost. We therefore define cost as negative similarity:

$$C_{ij} = -q_i^\top k_j.$$

The total cost of a plan  $P$  is

$$\sum_{i=1}^n \sum_{j=1}^m P_{ij} C_{ij} = - \sum_{i=1}^n \sum_{j=1}^m P_{ij} q_i^\top k_j,$$

so minimizing cost is the same as maximizing the similarity-weighted sum.

To entropically regularize the problem, we use the matrix Shannon entropy

$$H(P) = - \sum_{i=1}^n \sum_{j=1}^m P_{ij} \log P_{ij},$$

with the convention  $0 \log 0 = 0$ . We scale the negative entropy by a positive parameter  $\varepsilon$  and choose  $\varepsilon$  to match the attention temperature,  $\varepsilon = \tau = \sqrt{d_k}$ .

## *Objective and constrained problem*

Combine cost and negative entropy to form

$$J(P) = \sum_{i=1}^n \sum_{j=1}^m P_{ij} C_{ij} - \varepsilon H(P).$$

Substituting the expressions for cost and entropy gives

$$J(P) = - \sum_{i=1}^n \sum_{j=1}^m P_{ij} q_i^\top k_j + \varepsilon \sum_{i=1}^n \sum_{j=1}^m P_{ij} \log P_{ij}.$$

The generalized, one-sided entropic optimal transport problem that matches full scaled dot-product attention is

$$\min_{P \in U(a, \cdot)} J(P) \quad \text{with } \varepsilon = \tau = \sqrt{d_k}.$$

## *Existence and Uniqueness (theorem only)*

Let the source be  $1, \dots, n$  and the target be  $1, \dots, m$ . Let  $q_i$  and  $k_j$  be finite vectors in the common key dimension. Let the feasible set be

$$\mathcal{U} = \left\{ P \in \mathbb{R}_+^{n \times m} \mid \sum_{j=1}^m P_{ij} = 1 \right\}.$$

Here too, the row-sum constraint holds for all  $i$ .

For any  $\varepsilon > 0$ , the functional

$$J(P) = - \sum_{i=1}^n \sum_{j=1}^m P_{ij} q_i^\top k_j + \varepsilon \sum_{i=1}^n \sum_{j=1}^m P_{ij} \log P_{ij}$$

attains a unique minimum  $P^*$  on  $\mathcal{U}$ .

### *EOT solution equals the attention matrix*

Having the existence and uniqueness of the minimizer for the generalized entropic optimal transport problem over all queries and keys, we now derive its analytical form and show it equals the scaled dot-product attention matrix.

Let  $A = (A_{ij})$  be the attention matrix with temperature  $\tau = \sqrt{d_k}$ :

$$A_{ij} = \frac{\exp(q_i^\top k_j / \tau)}{\sum_{\ell=1}^m \exp(q_i^\top k_\ell / \tau)}, i = 1, \dots, n, j = 1, \dots, m.$$

Consider the one-sided entropic OT problem

$$\min_{P \in \mathcal{U}} J(P) = - \sum_{i=1}^n \sum_{j=1}^m P_{ij} q_i^\top k_j + \varepsilon \sum_{i=1}^n \sum_{j=1}^m P_{ij} \log P_{ij},$$

with  $\varepsilon = \tau$ . The unique minimizer  $P^*$  equals  $A$ .

### *Proof*

We minimize  $J(P)$  subject to the  $n$  equality constraints  $\sum_{j=1}^m P_{ij} = 1$  for each  $i$ , with the nonnegativity constraints  $P_{ij} \geq 0$ . Introduce Lagrange multipliers  $\lambda_1, \dots, \lambda_n$  for the row-sum constraints and write the Lagrangian



$$\mathcal{L}(P, \lambda) = - \sum_{i=1}^n \sum_{j=1}^m P_{ij} q_i^\top k_j + \varepsilon \sum_{i=1}^n \sum_{j=1}^m P_{ij} \log P_{ij} + \sum_{i=1}^n \lambda_i \left( 1 - \sum_{j=1}^m P_{ij} \right). \rightarrow$$

Take the partial derivative with respect to an arbitrary  $P_{ij}$  and set it to zero at the optimum. For any optimal solution with strictly positive entries (which will hold here), the stationarity condition is

$$- q_i^\top k_j + \varepsilon (\log P_{ij} + 1) - \lambda_i = 0.$$

Solve for  $P_{ij}$ :

$$\varepsilon \log P_{ij} = q_i^\top k_j + \lambda_i - \varepsilon.$$

$$\log P_{ij} = \frac{q_i^\top k_j + \lambda_i}{\varepsilon} - 1.$$

$$P_{ij} = \exp \left( \frac{\lambda_i}{\varepsilon} - 1 \right) \exp \left( \frac{q_i^\top k_j}{\varepsilon} \right).$$

Let  $Z_i = \exp \left( \frac{\lambda_i}{\varepsilon} - 1 \right)$ , which depends only on the row index  $i$ . Then

$$P_{ij} = Z_i \exp \left( \frac{q_i^\top k_j}{\varepsilon} \right).$$

Enforce the row-sum constraint to determine  $Z_i$ :

$$1 = \sum_{j=1}^m P_{ij} = Z_i \sum_{j=1}^m \exp \left( \frac{q_i^\top k_j}{\varepsilon} \right).$$

$$Z_i = \left[ \sum_{\ell=1}^m \exp \left( \frac{q_i^\top k_\ell}{\varepsilon} \right) \right]^{-1}.$$

Substitute back:

$$P_{ij} = \frac{\exp\left(\frac{q_i^\top k_j}{\varepsilon}\right)}{\sum_{\ell=1}^m \exp\left(\frac{q_i^\top k_\ell}{\varepsilon}\right)}.$$

Set  $\varepsilon = \tau$  and obtain

$$P_{ij} = \frac{\exp\left(\frac{q_i^\top k_j}{\tau}\right)}{\sum_{\ell=1}^m \exp\left(\frac{q_i^\top k_\ell}{\tau}\right)} = A_{ij}.$$

So the unique minimizer  $P^\star$  equals the scaled dot-product attention matrix  $A$ .

Positivity holds because exponentials are positive and the denominator is positive as long as dot products are finite. The solution lies in the relative interior and satisfies the KKT conditions. Uniqueness follows from strict convexity of  $J(P)$  on the convex feasible set.

## *The Discovery*

I should also mention that this is not, in the strictest sense, a full transport problem. In classical optimal transport, you fix both where the mass starts and where it ends up. But here, we only constrain the sources. Each query emits exactly one unit (a Dirac unit impulse) of attention, and we let the optimization decide how that mass gets distributed across the keys. There are no constraints on the target side. This makes it a one-sided transport problem, which turns out to be exactly what attention needs. If we had fixed both marginals, we would have gotten something else entirely, something that wouldn't match the softmax structure at all.

Memory is unreliable, especially when it comes to invention. I don't remember exactly the precise *Eureka!* moment of stumbling across this equivalence, though I suspect tangential work on Sinkhorn attention<sup>4</sup> warmed me up. Looking back, I can see the proof clearly.

Apparently this is not uncommon. It reminds me of Unix. Brian Kernighan admits he coined "Unics" as a pun during the early Bell Labs days, but even he and his collaborators couldn't determine exactly who decided on the final spelling "Unix." Ritchie and McIlroy credited Kernighan; Peter Neumann recalled that Bell Labs PR likely influenced the change to avoid the "eunuchs" joke. No one remembers.  
([Wikipedia](#))

What happens is that thinking is computationally expensive: you explore dead ends, trace connections that lead nowhere, hold contradictory possibilities simultaneously. When you arrive at a solution, you amortize the cost by compressing the search path into the final answer and discarding intermediate states. This is your mind pruning its search history after convergence, like AlphaGo discarding its tree search after the move is chosen. The moment you see it, the path to seeing it disappears. Once the connection is made, the confusion that preceded it becomes incomprehensible because the information-theoretic cost of preserving that confusion exceeds any utility it might have held. ■

Finalized at 7:26 AM.

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